mise

Cursor for restaurants

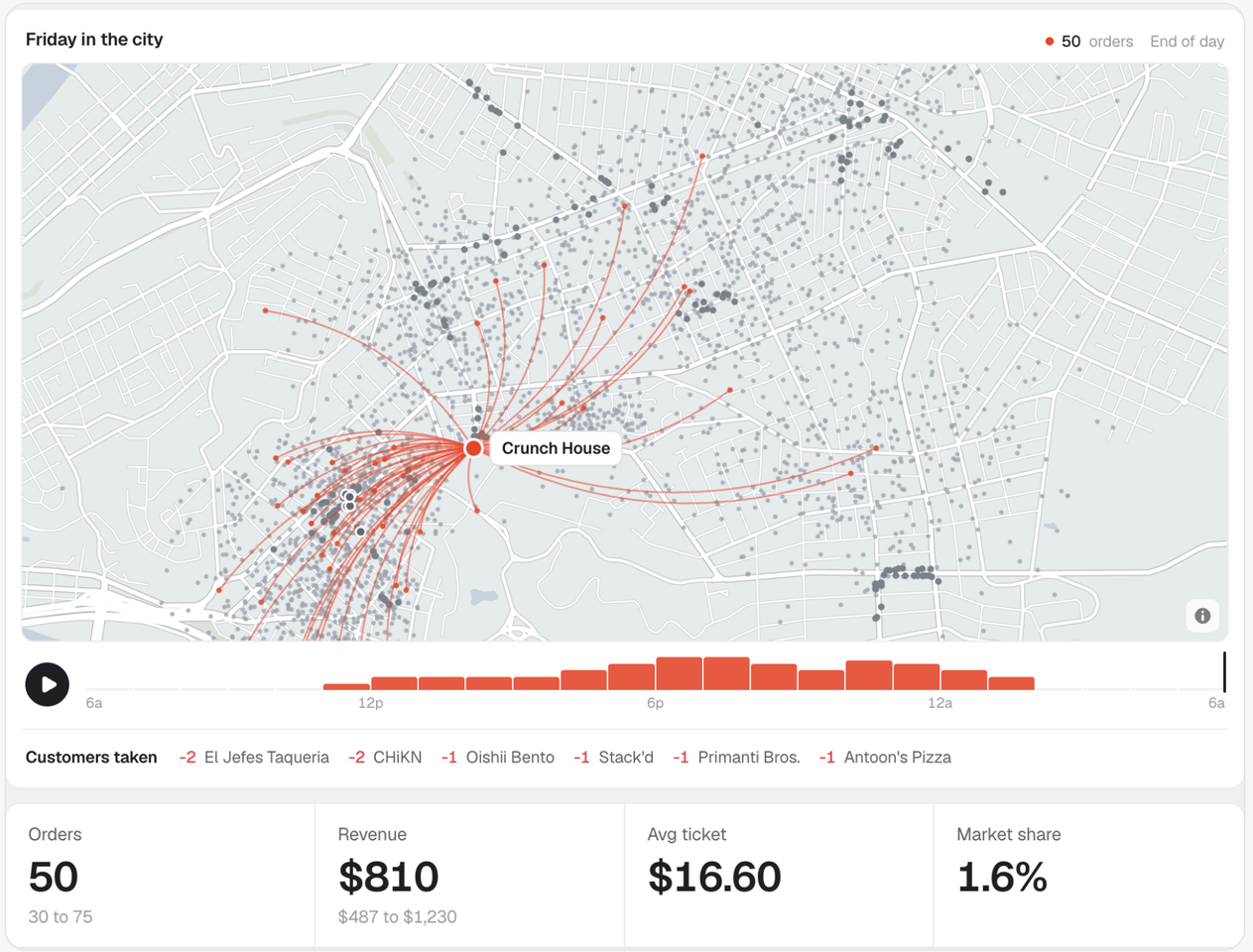
Test a restaurant idea on a simulated city before you open it.

Team Peanuts

Zhiyuan Guo, Evelyn Wang

Track

DoorDash, Back House



A new restaurant's first Friday, before it exists. 10,000 simulated residents, 50 predicted orders.

The problem

Owners can only test a change by making it

3–5%

typical restaurant margin

Little room for a bad bet.

Source: DoorDash

Decision	How they find out today	If it's wrong
Open a new concept	Sign a lease, open, wait	Savings and a lease
Change prices	Change the menu, watch for weeks	Lost customers or margin
Lower the delivery fee	Run it, check the payouts	Fees given away
Stay open later	Schedule staff, see who comes	Paid hours, no orders

Mise lets them try it on a simulated city first.

Who it's for

Two kinds of owners

Opening a new place

Has a concept and a location in mind. Doesn't know who will order, at what hours, or what to charge.

Wants to know

Is there enough demand here, and who would I take customers from?

Already open

Thinking about a lower delivery fee or staying open later. Every real test costs a week of revenue.

Wants to know

Who does the change bring in, and is it worth what it costs?

How it works

01

Describe the idea in chat, or start from a real restaurant

02

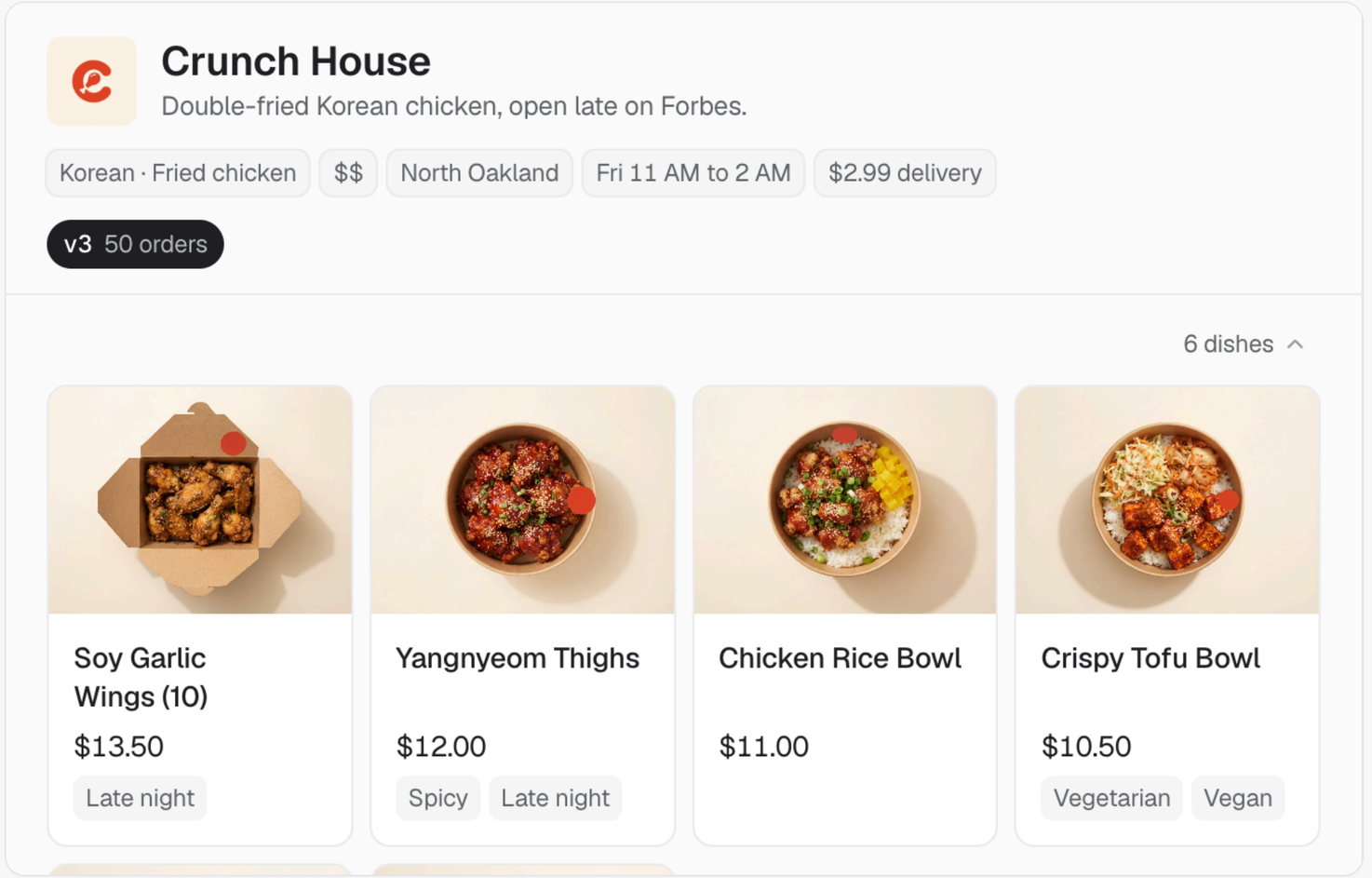
Mise drafts it: name, menu, prices, logo, dish photos

03

Simulate a day in the city, hour by hour

04

Set a goal and approve the edits that get there

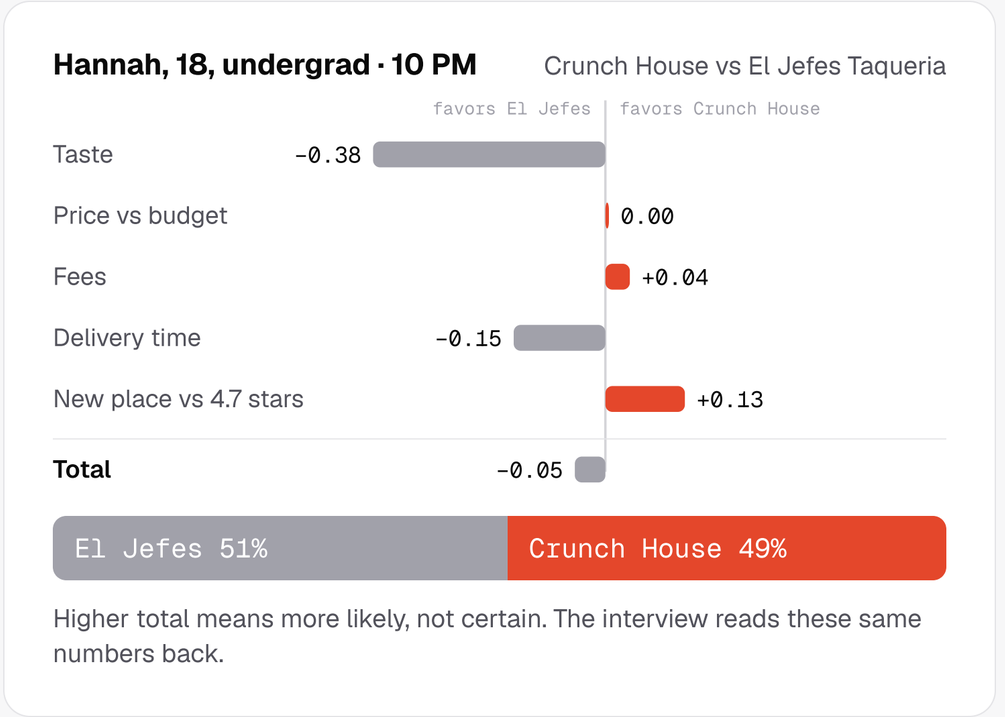


Live demo

What each resident weighs

Each hungry resident scores every open place within 3 km, plus cooking or skipping.

Food	**Taste**	their cuisines, by time of day
	Diet options	needs something they can eat
Money	**Price vs budget**	over budget hurts, under helps a little
	Fees	vs usual $2.99 + 14%, hurt 1.2x more
	Deals	launch discount gets people to try
Speed	**Delivery time**	18 min + 5 min per km
Trust	**Rating**	stars above or below 4.0
	Reviews	more reviews, picked more
	New place	unknown places start behind
	Habit	favorite places get a boost



Higher total means more likely, not certain. The interview reads these same numbers back.

Accuracy

How we know the numbers hold up

Matches real restaurants	Fit to Google busy times for 51 places, tested on 13 it never saw	r 0.54 vs 0.44 baseline
Behaves like a city	Automatic checks for direction, double counting and consistency	26 of 26 pass
Stable headline	100 simulated Fridays per prediction instead of 20	spread 12 → 6 orders
Fair before and after	Every version faces the same residents and the same random draws	only the edit differs
Optimizer re-checked	Final edits re-scored on simulated days the search never saw	beat original 10 of 10
Model proposes, code decides	Edits are clamped by code, scored by the simulator, approved by the owner	predictions never from the LLM

Accuracy

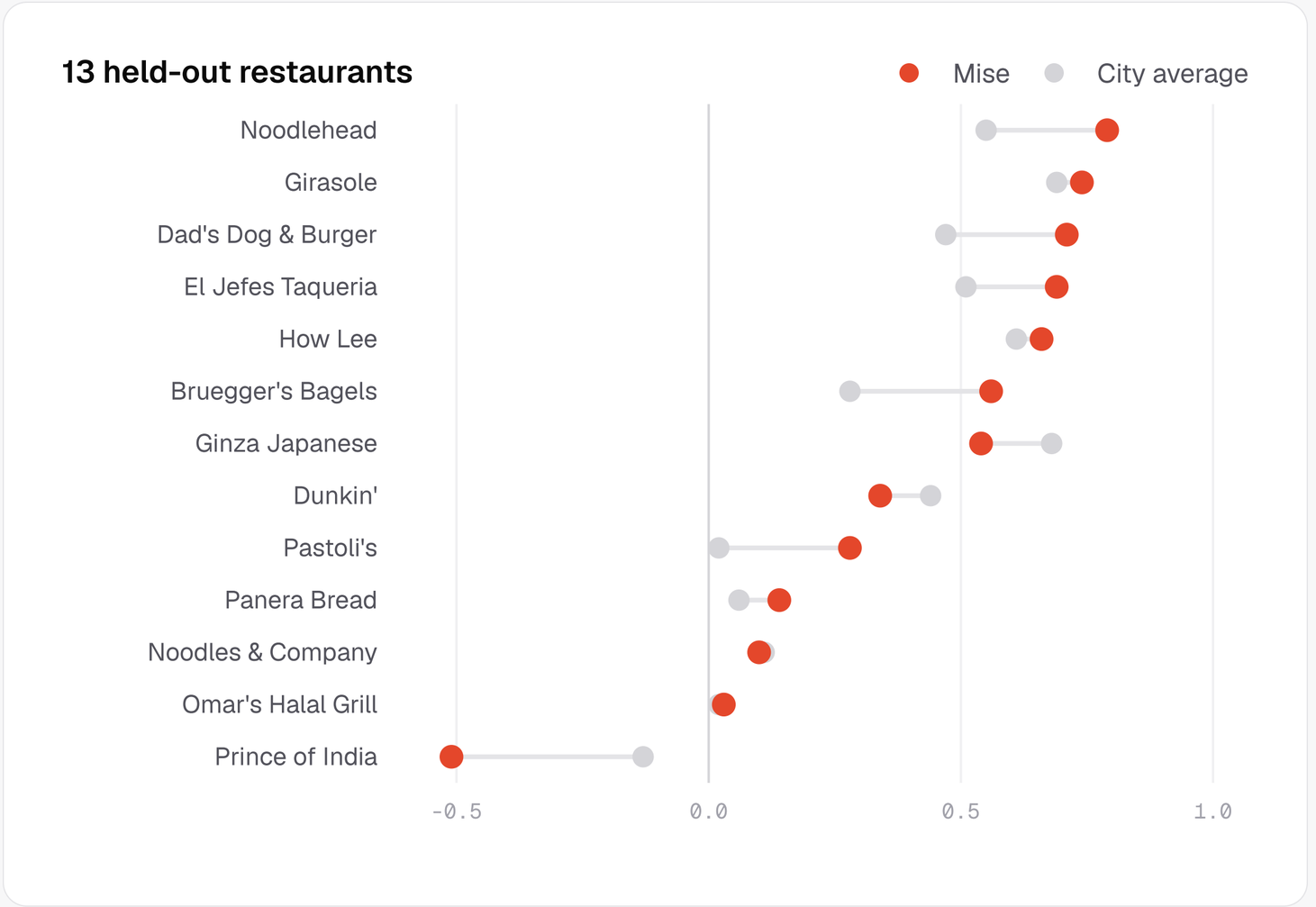
It matches real restaurants it never saw

0.54 Mise, calibrated

0.44 City-average curve for every place

0.32 Mise before calibration

Median correlation with each place's real hourly curve. We fit on 51 restaurants and tested on 13 the model never saw.



Accuracy

26 checks run on every change

Direction

Cheaper means more orders. Pricier, farther, or closing early means fewer. Free delivery and a launch promo mean more. Monday is below Friday. A vegan main cuts diet objections.

Words can't move numbers

Editing only the tagline gives identical orders.

One story

For sampled residents, the list, the map and the interview agree on what they ordered.

Import a real restaurant unchanged

It should get back its own orders and take nothing from itself.

	Today in the sim	Imported as-is	Taken from itself
Noodlehead	46	45	0
CHiKN	66	69	0
Pizza Palermo	53	55	0
Sushi Fuku	9	10	0

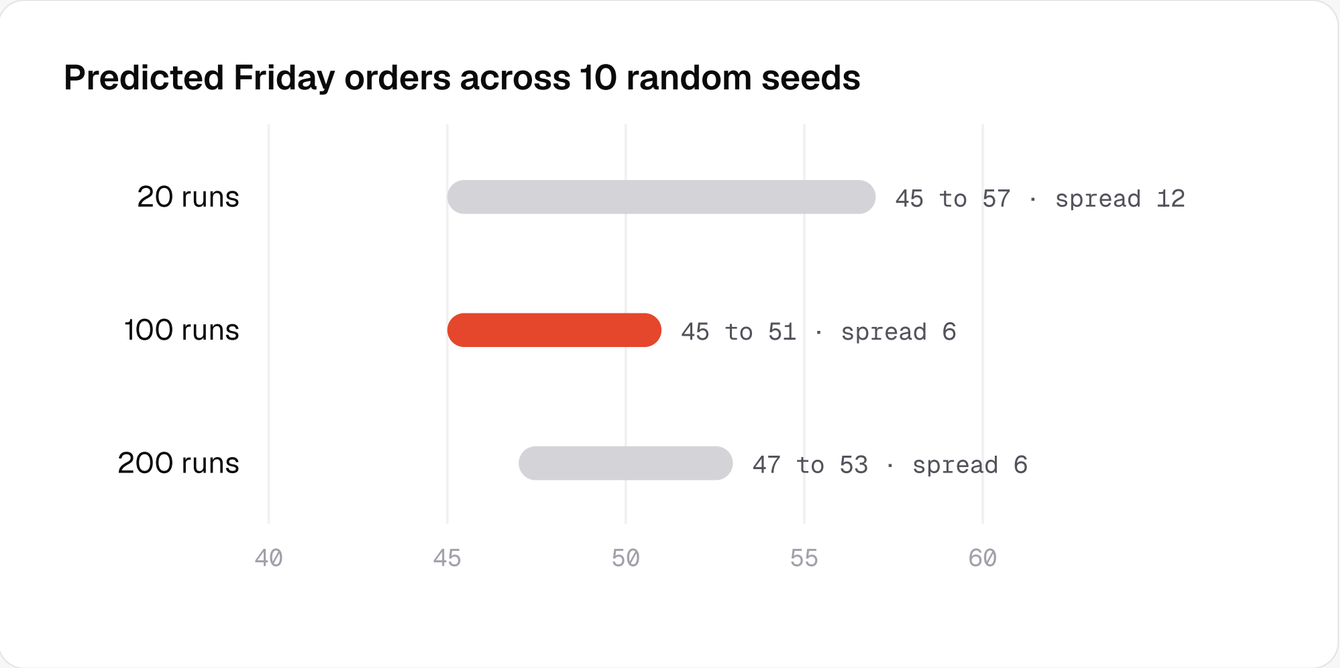
Accuracy

Enough runs that the headline doesn't wobble

Each prediction simulates many Fridays. Every run draws a city-wide demand level, how fast people find a new place, and every resident's orders.

At 20 runs, the headline moved by up to **12 orders** between random seeds. At 100 runs it moves by **6**, and 200 runs doesn't improve on that.

So every prediction uses 100 runs, about 1.4 s.



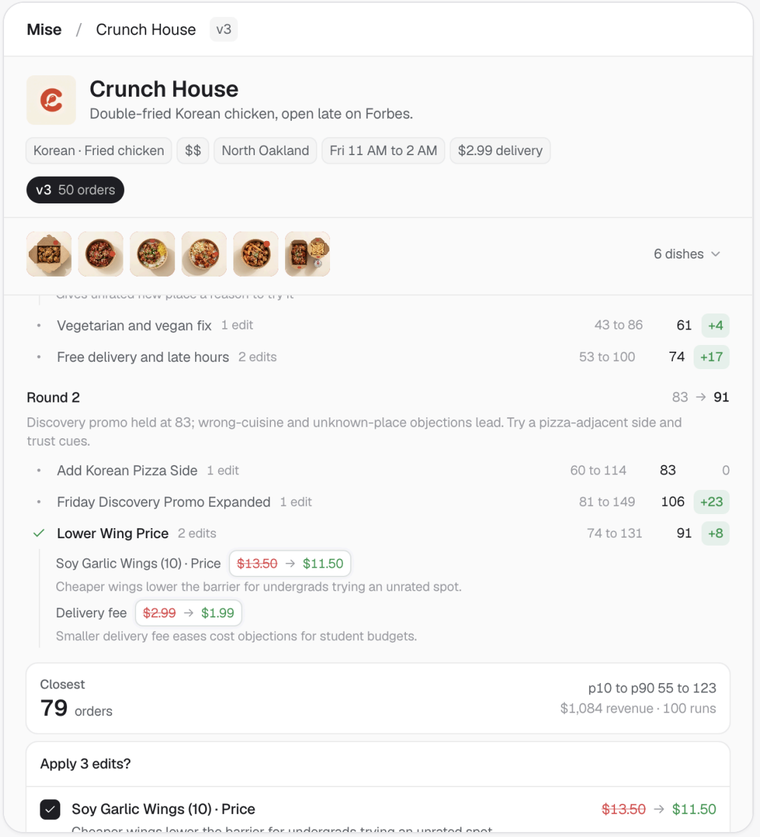
Accuracy

The optimizer is checked on days it never saw

The search tries edit sets and keeps the best score. The best of several noisy scores runs high. So the number the owner sees comes from a fresh check: 100 new Fridays, original and edited concept on the same days.

Search's own score	91 orders
Fresh check, shown to the owner	79 orders
10 more fresh seeds, beat the original	10 of 10
Median across those seeds	49.5 → 83

Goal: 90 orders on Friday. Every edit has limits: 3 per set, no dish under 60% of the owner's price, discounts up to 30%, fee up to $5.99. Nothing applies until the owner approves.



Trade-offs

What's real and what's estimated

Real

192 restaurants from OpenStreetMap and Google Maps

Google popular times for 64 of them

Menus pulled from the restaurant's own site when every item and price is on the page

Estimated

The resident mix per neighborhood is hand-set. Census data is next

Google shows how busy a place is, not order counts, so we match the shape of the day, not volume

Price and fee sensitivity are hand-set from typical delivery-app behavior, not fitted

Menus without a website are drafted and labeled that way

Stack TypeScript, Bun + Hono, React, MapLibre + deck.gl. Claude Sonnet for chat, Claude Haiku for the goal loop and interviews, Gemini for images, through OpenRouter.

With DoorDash

What this becomes with real order data

Calibrate on real orders	Swap Google's busy curves for each merchant's order history, so the model learns volume and price sensitivity, not only timing.
Any city	Pittsburgh is built. Another city needs its restaurant list, busy times and resident mix.
Test before you change it	An owner tries a price, hours or fee change in the merchant tools and sees the result before turning it on.

Thanks!

github.com/ZhYGuoL/mise